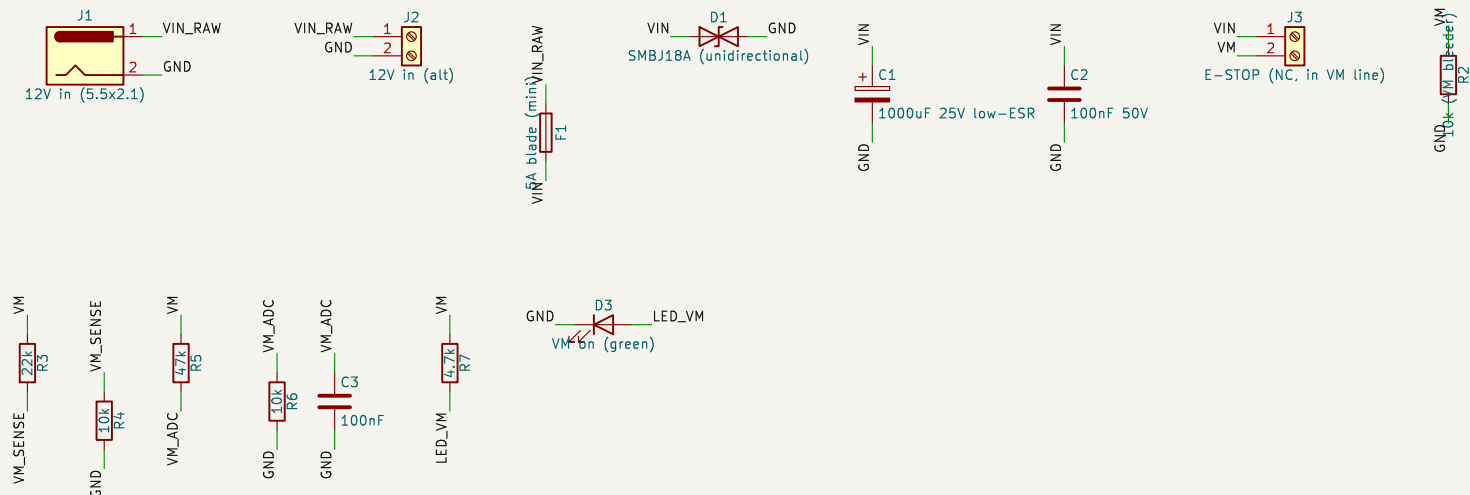
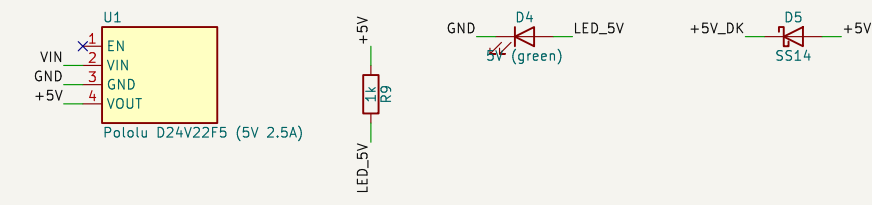


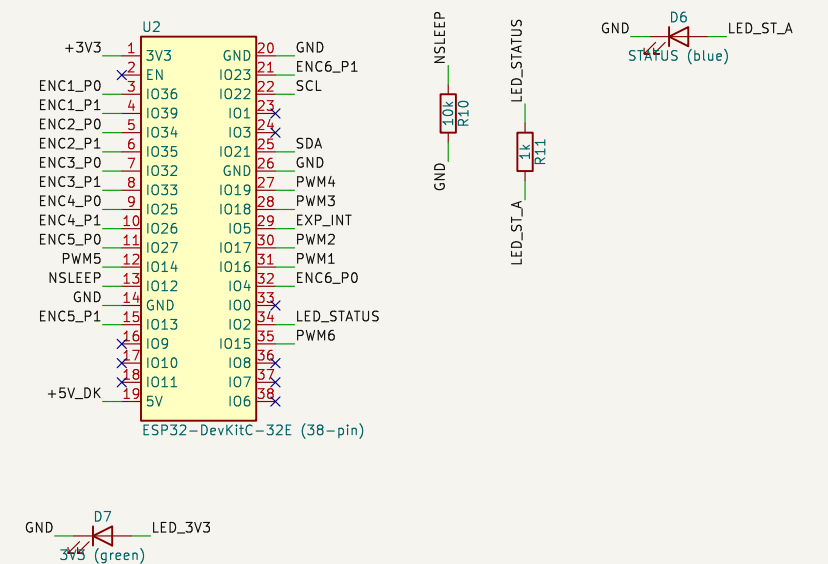
POWER IN, REVERSE PROTECTION, E-STOP (no relay; NC contacts in VM line)



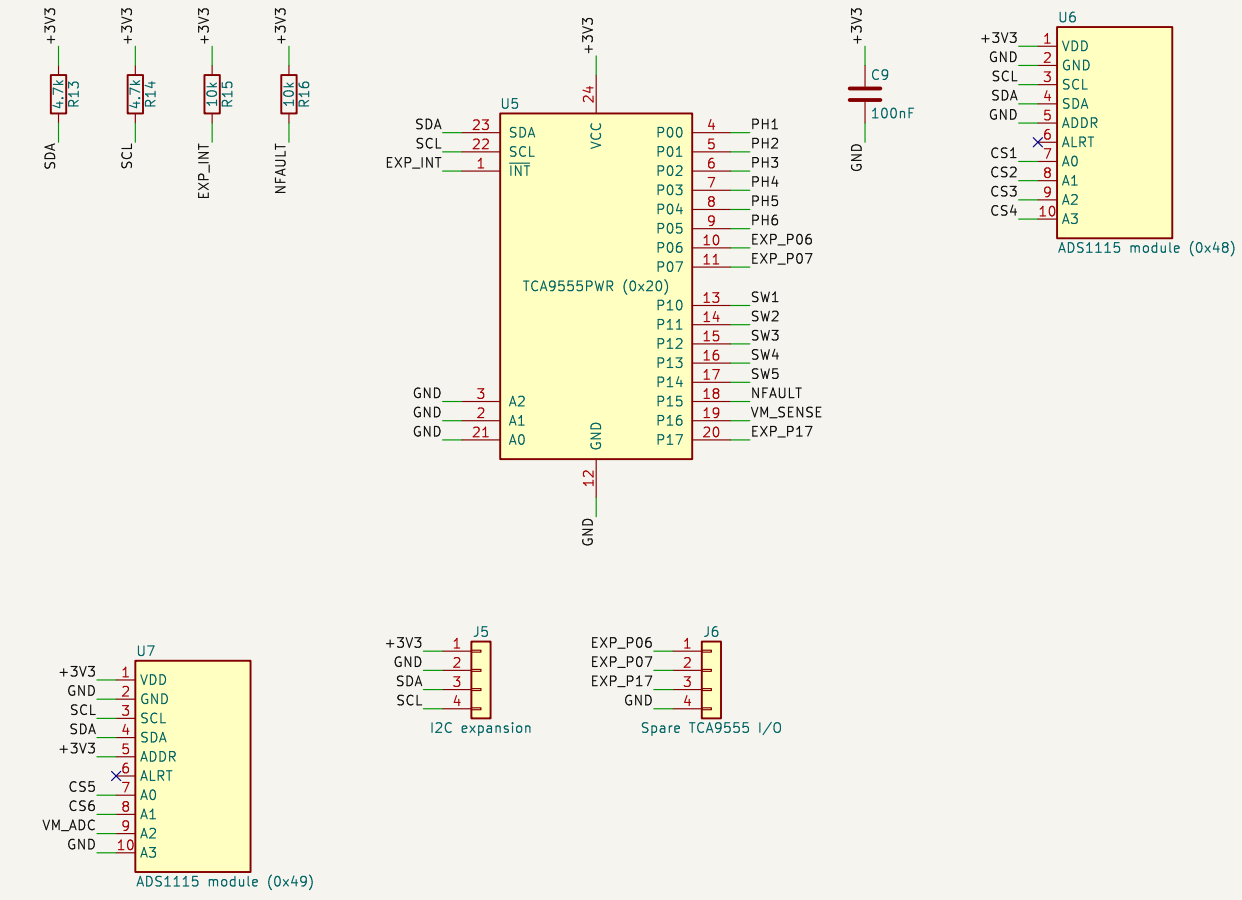
5 V: Pololu D24V22F5 plug-in buck + DEVKIT 5V FEED



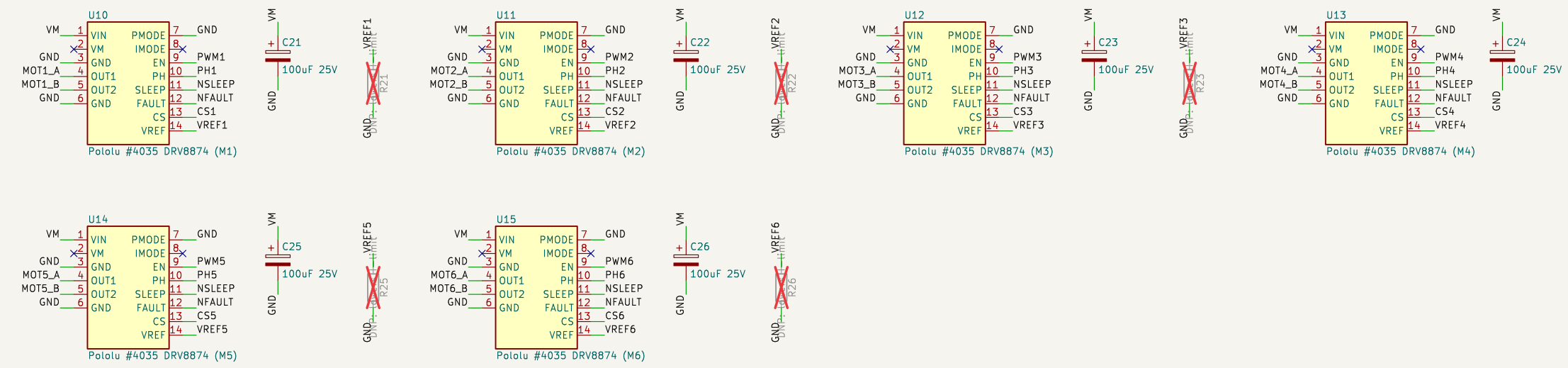
ESP32-DevKITC-32E (plug-in, 2x 1x19 female headers)



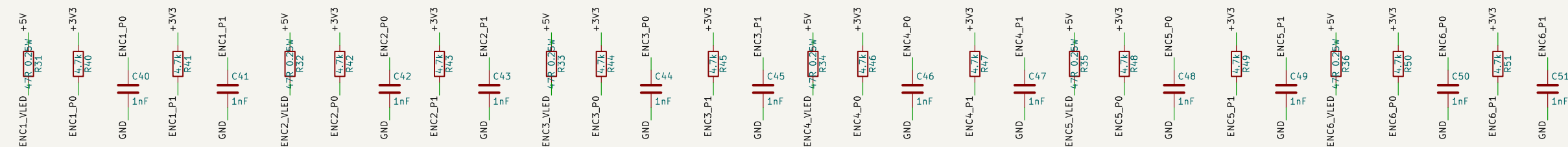
I2C: TCA9555 (0x20) + 2x ADS1115 modules (0x48, 0x49)



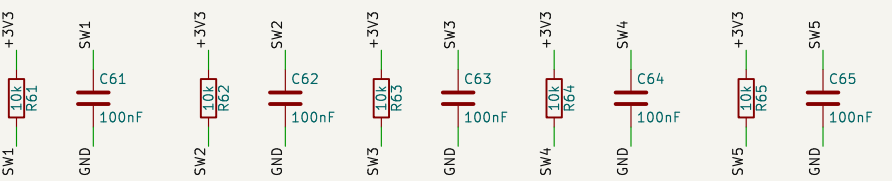
MOTOR DRIVERS: 6x Pololu DRV8874 carrier (plug-in) – PIN ORDER PLACEHOLDER



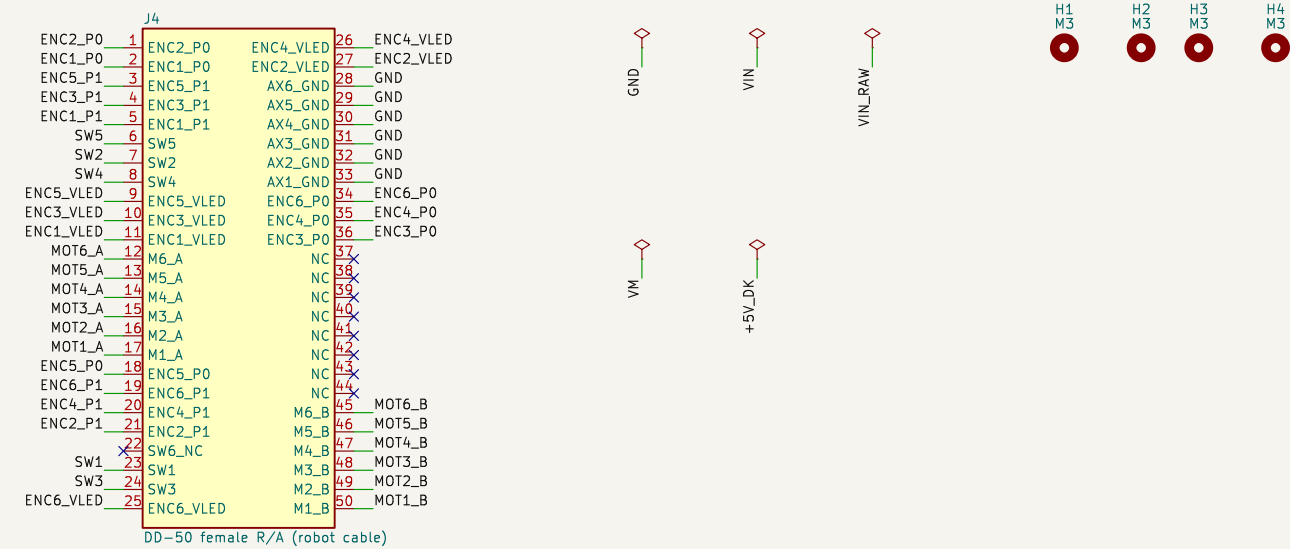
ENCODER FRONT END: 47R VLED from 5V, 4.7k pull-ups to 3.3V, 1nF, direct to ESP32



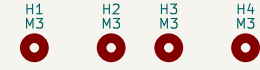
HOME SWITCHES (to TCA9555 P1.0–P1.4)



ROBOT CABLE: DD–50 female (SCORBOT–ER 4u manual R0WB09 FLAGS)



MOUNTING



- NOTES
1. Pololu DRV8874 carrier pin ORDER is a placeholder (names are correct). Fix design.CARRIER_ROWS and regenerate.
 2. DB50 pinout from SCORBOT–ER 4u User Manual ch. 8 (same as ER III Table D–1). Pins 37–44 unused, 22 = gripper switch (no connection in arm).
 3. Plug-in module pin orders (Pololu D24V22F5, ADS1115 breakout) are placeholders; check against your modules.
 4. Carrier current limit with SLEEP at 3.3 V is about 2.9 A (VREF pulled to SLEEP by 10k on the carrier, CS has 2.49k). R21–R26 (DNP) lower it.
 5. Current sense: CS reads 1.12 V/A into ADS1115 modules (use the +/-4.096 V range). VM monitored via 47k/10k on ADS1115 #2 A2.